

NCERT Exemplar Solutions for Class 12

Chemistry

Chapter 13 - Amines

Multiple Choice Questions (Type-I)

1. Which of the following is a 3° amine?

(A) 1-methylcyclohexylamine

(B) Triethylamine

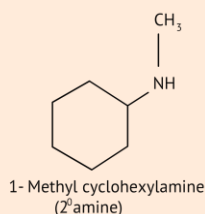
(C) tert-butylamine

(D) N-methylaniline

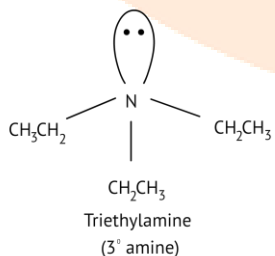
Ans: Correct answer is (B) Triethylamine

The structure of the given amines are shown below.

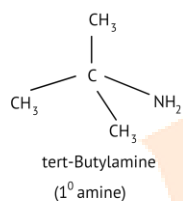
(A)



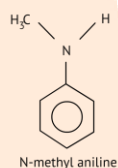
(B)



(C)



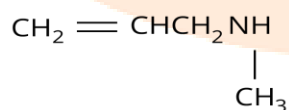
(D)



2. The correct IUPAC name for $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{NH} - \text{CH}_3$ is

- (A) allylmethylamine
- (B) 2-amino-4-pentene
- (C) 4-aminopent-1-ene
- (D) N-méthylprop-2-en-1-amine

Ans: Correct answer: (D)



3. Amongst the following, the strongest base in an aqueous medium is.



Ans: Correct answer is (C)

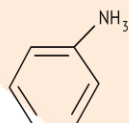
Because of the electron-releasing nature of the alkyl group, it (R) pushes electrons towards nitrogen, making the unshared electron pair more available for sharing with the acid's proton. Furthermore, the +I effect of the alkyl group stabilizes the substituted ammonium ion formed from the amine by dispersing the positive charge.

As a result, alkylamines are more powerful bases than ammonia.

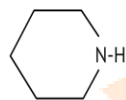
As a result, the basic nature of aliphatic amines should increase as the number of alkyl groups increases.

4. Which of the following is the weakest Bronsted base?

(A)



(B)



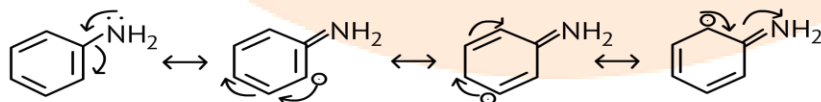
(C)



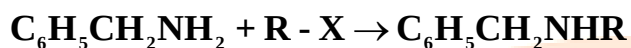
(D) CH_3NH_2

Ans: Correct answer is (A)

The NH_2 group is directly attached to the benzene ring in aniline and other aryl amines. As a result, the unshared electron pair on the nitrogen atom is conjugated with the benzene ring, making it less available for protonation.



5. Benzylamine may be alkylated as shown in the following equation:



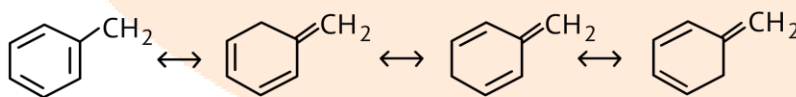
Which of the following alkyl halides is best suited for this reaction through SN^1 mechanism?

- (A) CH_3Br
- (B) $\text{C}_6\text{H}_5\text{Br}$
- (C) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$
- (D) $\text{C}_2\text{H}_5\text{Br}$

Ans: Correct answer is (C)

SN^1 reaction: A nucleophilic reaction that occurs in two steps, first is the bond-breaking step and the second is the production of the carbocation. The stability of carbocation formed in the second step determines the rate of reactivity of reactant toward SN^1 reaction. Here, $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$,

In the process of ionization, removal of bromine, a stable Benzyl carbocation is produced. Therefore, it is best suited for reaction through the SN^1 mechanism.

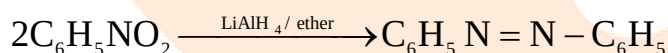
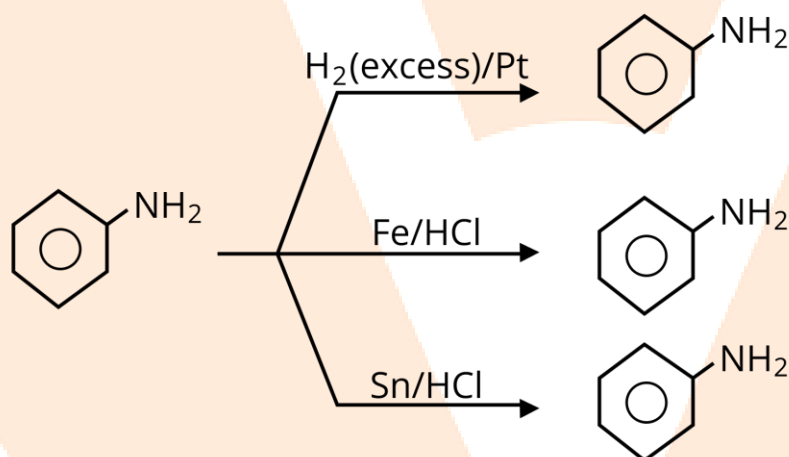


6. Which of the following reagents would not be a good choice for reducing an aryl nitro compound to an amine?

- (A) H_2 (excess)/Pt
- (B) LiAlH_4 in ether
- (C) Fe and HCl
- (D) Sn and HCl

Ans: Correct answer is (B)

Lithium aluminum hydride in ether is a strong reducing agent that donates its H^- hydride ion to any $\text{C}=\text{O}$ containing a functional group. In addition to LiAlH_4 to aryl nitro compounds, no reaction will be observed, the desired products of amines will not be produced, rather it will form diazobenzene products.



7. In order to prepare a primary amine from an alkyl halide with simultaneous addition of one CH_2 group in the carbon chain, the reagent used as a source of nitrogen is

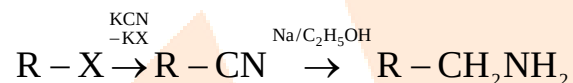
- (A) sodium amide, NaNH_2
- (B) sodium azide, NaN_3

(C) potassium cyanide, KCN

(D) potassium phthalimide, $C_6H_4(CO_2)N^-K^+$

Ans: Correct answer is (C)

Potassium cyanide, as cyanide on reduction with sodium metal in alcohol, produces amine with increased carbon atoms.



8. The source of nitrogen in Gabriel synthesis of amines is.

(A) sodium azide, NaN_3

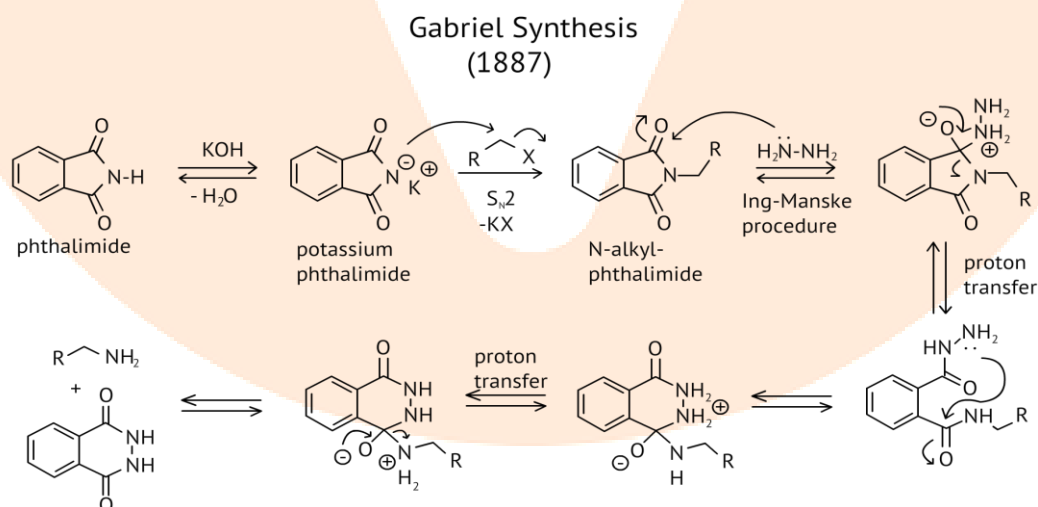
(B) sodium nitrite, $NaNO_2$

(C) potassium cyanide, KCN

(D) potassium phthalimide, $C_6H_4(CO_2)N^-K^+$

Ans: Correct answer is (D)

Gabriel synthesis reaction:



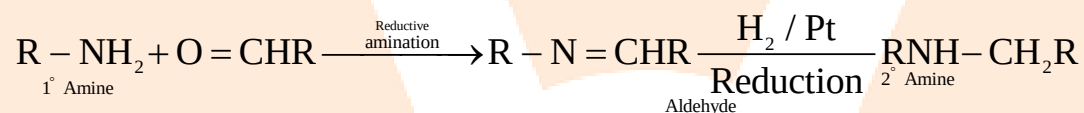
Gabriel's synthesis is a method for producing primary amines. When phthalimide is treated with ethanolic potassium hydroxide, it forms a potassium salt of phthalimide,

which when heated with an alkyl halide and then alkaline hydrolyzed yields the corresponding primary amine. This method cannot produce aromatic primary amines because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.

9. Amongst the given set of reactants, the most appropriate for preparing 2° amine is.

- (A) 2° R - Br + NH₃
- (B) 2° R - Br + NaCN followed by H₂ / Pt
- (C) 1° R - NH₂ + RCHO followed by H₂ / Pt
- (D) 1° R - Br + (2 mol) + potassium phthalimide followed by H₃O⁺ / heat

Ans: Correct answer is (C)



10. The best reagent for converting 2-phenylpropanamide into 2-phenylpropanamine is.

- (A) excess H₂
- (B) Br₂ in aqueous NaOH
- (C) iodine in the presence of red phosphorus
- (D) LiAlH₄ in ether

Ans: Correct answer is (D)

Lithium aluminium hydride in ether is a strong reducing agent that donates its hydride ion to any C = O containing functional groups into corresponding reduced compounds. Here, the amide group is converted to an amine functional group.

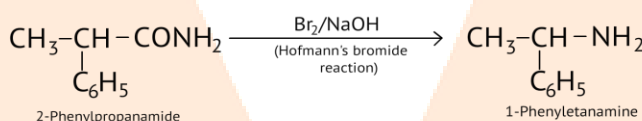
LiAlH_4 in ether is the best reagent for converting 2-phenylpropanamide to 2-phenylpropanamine.

11. The best reagent for converting-2-phenylpro-panamide into 1-phenylethanamine is.

- (A) excess H_2 / Pt
- (B) NaOH / Br_2
- (C) NaBH_4 / methanol
- (D) LiAlH_4 / ether

Ans: Correct answer is (B)

Hoffmann bromamide reaction- it is also called degradation reaction as in this reaction primary amide group is treated with halogen first (Br) then the halogen-substituted amide product is converted to a primary amine with the release of carbon dioxide gas.



12. Hoffmann bromamide degradation reaction is shown by.

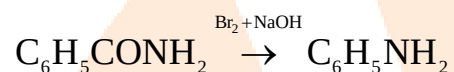
- (A) ArNH_2
- (B) ArCONH_2

(C) ArNO_2

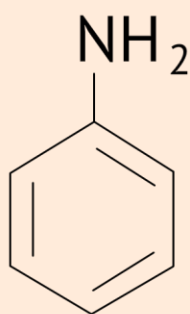
(D) ArCH_2NH_2

Ans: Correct answer is (B)

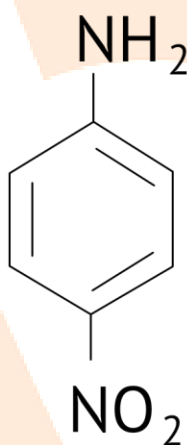
Hoffmann invented a method for producing primary amines by treating an amide with bromine in an aqueous or ethanolic sodium hydroxide solution. An alkyl or aryl group migrates from the amide's carbonyl carbon to the nitrogen atom during this degradation reaction.



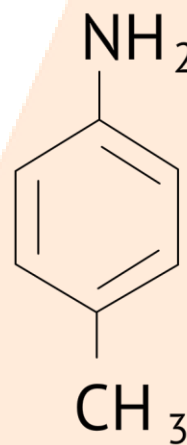
13. The correct increasing order of basic strength for the following compounds is



(i)



(ii)



(iii)

(A) $\text{II} < \text{III} < \text{I}$

(B) $\text{III} < \text{I} < \text{II}$

(C) $\text{III} < \text{II} < \text{I}$

(D) $\text{II} < \text{I} < \text{III}$

Ans: Correct answer is (D)

The greater the electron density towards the ring, the greater its basic strength.

The electron withdrawing group reduces basic strength, whereas the electron donating group increases basic strength.

14. Methylamine reacts with HNO_3 to form

(A) $\text{CH}_3 - \text{O} - \text{N} = \text{O}$

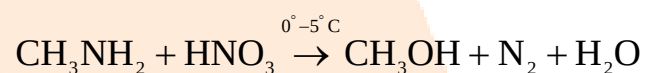
(B) $\text{CH}_3 - \text{O} - \text{CH}_3$

(C) CH_2OH

(D) CH_3CHO

Ans: Correct answer is (C)

Methyl amine reacts with HNO_3 form methanol with release of nitrogen gas and water as side products.



15. The gas evolved when methylamine reacts with nitrous acid is .

(A) NH_3

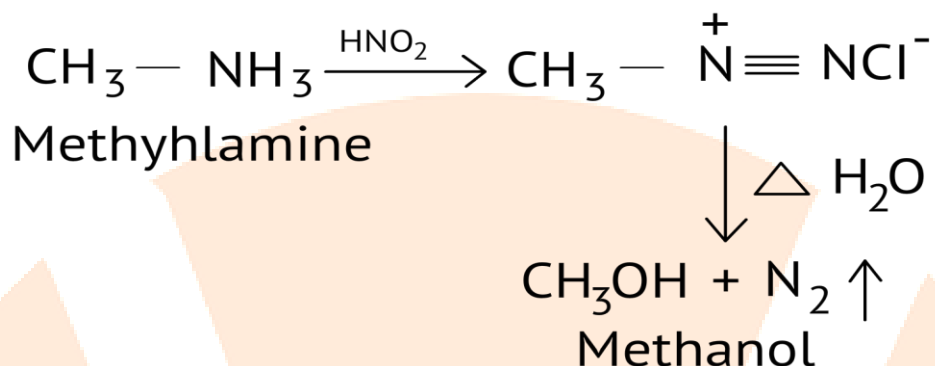
(B) N_2

(C) H_2

(D) C_2H_6

Ans: Correct answer is (B)

Primary aliphatic amines react with nitrous acid to form aliphatic diazonium salts, which are unstable and release nitrogen gas and alcohol quantitatively.

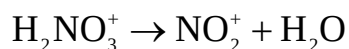
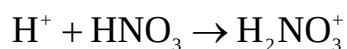
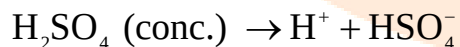


16. In the nitration of benzene using a mixture of conc. H_2SO_4 and conc. HNO_3 , the species which initiates the reaction is.

- (A) NO_2
- (B) NO^+
- (C) NO_2^+
- (D) NH_2

Ans: Correct answer is (C)

In the process of nitration of benzene, firstly H_2SO_4 dissociates into H^+ and HSO_4^- . The proton produced further reacts with HNO_3 to give a positively charged complex that is unstable and dissociates into nitronium ions and water as products. The nitronium ion is an electron-deficient species i.e. electrophile that attacks the electron cloud of the benzene ring.



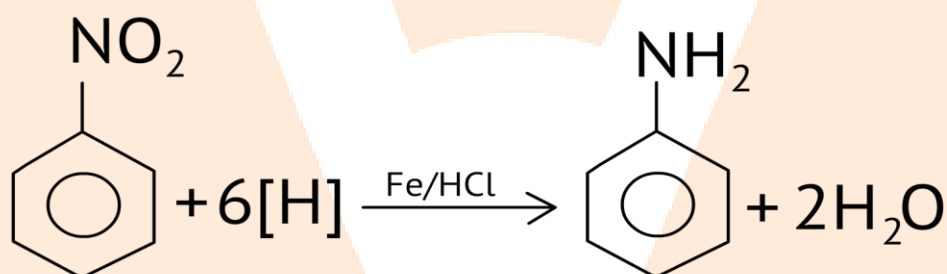
17. Reduction of aromatic nitro compounds using Fe and HCl gives.

- (A) aromatic oxime
- (B) aromatic hydrocarbon
- (C) aromatic primary amine
- (D) aromatic amide

Ans: Correct answer is (C)

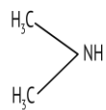
Using active metals like iron in acidic conditions can be efficiently used for the reduction of nitro compounds.

Under the acidic conditions, the intermediate compounds that may form are readily reduced to primary amine and pure product of primary amine is obtained.

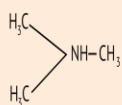


18. The most reactive amine towards dilute hydrochloric acid is

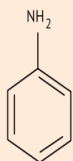
- (A) $\text{CH}_3 - \text{NH}_2$
- (B)



(C)



(D)



Ans: Correct answer is (B)

Greater will be the basic character, greater will be its reactivity towards dilute hydrochloric acid. As, (B), secondary amine, it will be the most basic among all and hence the most reactive amine towards acid.

19. Acid anhydrides on reaction with primary amines give.

(A) amide

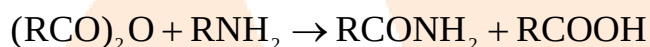
(B) imide

(C) secondary amine

(D) imine

Ans: Correct answer is (A)

On Reaction of acid anhydrides with primary amines, amides are formed as the primary product along with carboxylic acid.



20. The reaction as $\text{ArN}_2\text{Cl} \xrightarrow{\text{Cu/HCl}} \text{ArCl} + \text{N}_2 + \text{CuCl}$ is names

(A) Sandmeyer reaction

(B) Gatterman reaction

(C) Claisen reaction

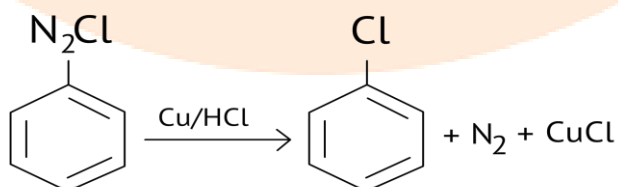
(D) Carbylamine reaction

Ans: Correct answer is (B)

Diazonium salts react with copper powder and halogen acid to form aryl halide.

Gattermann reaction is a Sandmeyer reaction variant in which Cu powder replaces CuCl.

This substitution produces aryl halide more easily and under milder conditions than the Sandmeyer reaction.

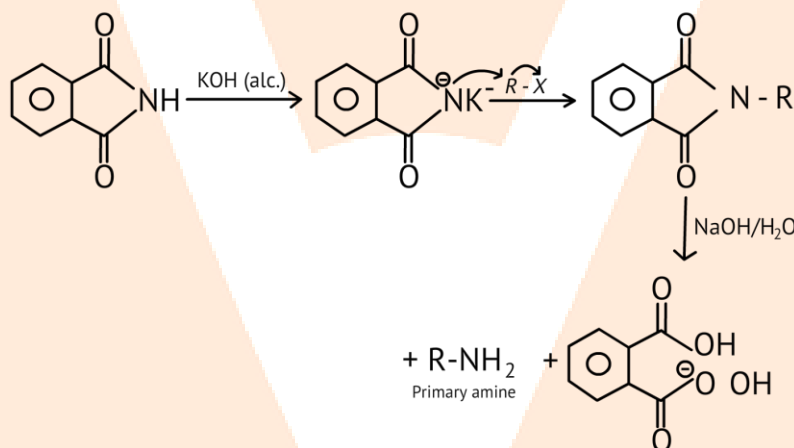


21. Best method for preparing primary amines from alkyl halides without changing the number of carbon atoms in the chain is

- (A) Hoffmann bromamide reaction
- (B) Gabriel phthalimide reaction
- (C) Sandmeyer reaction
- (D) reaction with NH_3

Ans: Correct answer is (B)

The reaction of transformation of primary alkyl halides to primary amines using potassium phthalimide is the Gabriel phthalimide reaction.



22. Which of the following compounds will not undergo azo coupling reaction with benzene diazonium chloride?

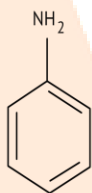
- (A) Aniline
- (B) Phenol
- (C) Anisole
- (D) Nitrobenzene

Ans: Correct answer is (D)

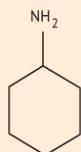
Weak electrophile such as Diazonium cation readily reacts with electron-rich compounds which are having electron-rich compounds such as hydroxyl group, amino group. They don't react with electron-withdrawing groups like nitro groups. Therefore, nitrobenzene will not undergo an azo coupling reaction with benzene diazonium chloride.

23. Which of the following compounds is the weakest Bronsted base?

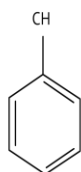
(A)



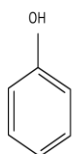
(B)



(C)



(D)



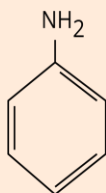
Ans: Correct answer is (D)

Since phenol is the strongest acid among the four options listed above, it is the weakest Brønsted base. The stronger the acid, the weaker the conjugate base.

Amines have a strong tendency to accept electrons thus they are a strong Brønsted base while phenol is the strongest acid among all, therefore as per the relation of conjugative strong acids and weak bases, phenol is the weakest base.

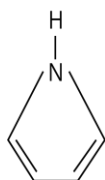
24. Among the following amines' the strongest Brønsted base is

(A)



(B) NH_3

(C)



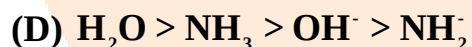
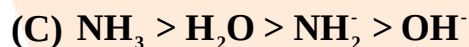
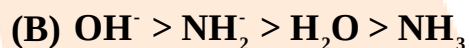
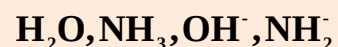
(D)



Ans: Correct answer is (D)

Pyrrolidine is the strongest of two bases because the lone pair of nitrogen does not involve in resonance, and the presence of two alkyl basic compounds increases the basic strength among the given four compounds.

25. The correct decreasing order of basic strength of the following species is



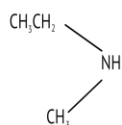
Ans: Correct answer is (A)

The basic strength of an atom is determined by its electron-donating capacity; in this case, the amide is the most basic due to the presence of a negative charge and two lone pairs of electrons on the nitrogen atom.

26. Which of the following should be most volatile?



(iii)



(iv) $\text{CH}_3\text{CH}_2\text{CH}_3$

(A) II

(B) IV

(C) I

(D) III

Ans: Correct answer is (B)

Among primary, secondary, and tertiary amines, tertiary amines are the most volatile compounds as they don't have any strong intermolecular H-bonding between N-H like the primary and secondary amines have.

So due to weak dipole-dipole interactions, (B) will be the most volatile.

27. Which of the following methods of preparation of amines will not give the same number of carbon atoms in the chain of amines as in the reactant?

(A) Reaction of nitrile with LiAlH_4

(B) Reaction of amide with LiAlH_4 followed by treatment with water.

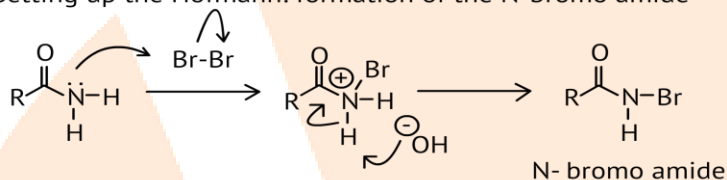
(C) Heating alkyl halide with potassium salt of phthalimide followed by hydrolysis

(D) Treatment of amide with bromine in aqueous solution of sodium hydroxide.

Ans: Correct answer is (D)

Only treatment of amide with Br_2 in aqueous NaOH solution yields an amine with fewer carbon atoms than the reactant, whereas $\text{RCONH}_2\text{Br}_2 / \text{NaOHRNH}_2$ all other reactions yield an amine with the same number of carbon atoms as the reactant.

Setting up the Hofmann: formation of the N-bromo amide



Multiple Choice Questions (Type- II)

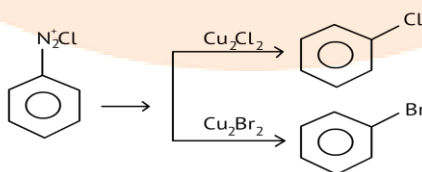
28. Which of the following cannot be prepared by Sandmeyer's reaction?

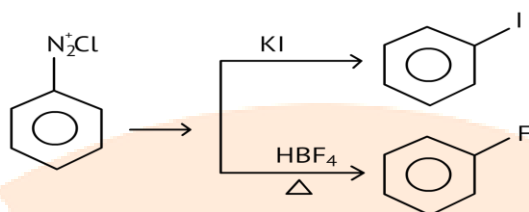
- (A) Chlorobenzene
- (B) Bromobenzene
- (C) Iodobenzene
- (D) Fluorobenzene

Ans: Correct answer is (C and D)

Chlorobenzene and bromobenzene are prepared by the sandmeyer's reaction.

While fluorobenzene and iodobenzene are prepared by simple heating of diazonium salt with aqueous KI Solution.





29. Reduction of nitrobenzene by which of the following reagents give aniline?

(A) Sn/HCl

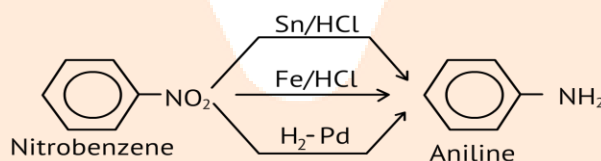
(B) Fe/HCl

(C) $\text{H}_2 - \text{Pd}$

(D) $\text{Sn} / \text{NH}_4\text{OH}$

Ans: Correct answer is (A, B & C)

Nitro compounds are reduced to amines by passing hydrogen gas through an acidic medium containing finely divided nickel, palladium, or platinum, as well as by reduction with metals in an acidic medium.



30. Which of the following species are involved in the carbylamine test?

(A) $\text{R}-\text{NC}$

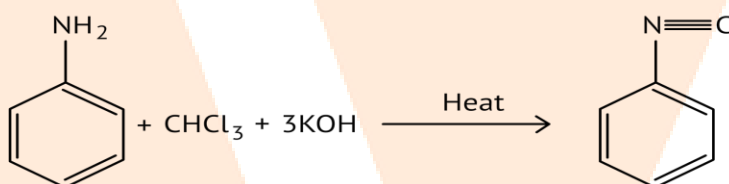
(B) CHCl_3

(C) COCl_2

(D) $\text{NaNO}_2 + \text{HCl}$

Ans: Correct answer is (A and B)

When aliphatic and aromatic primary amines are heated with chloroform and ethanolic potassium hydroxide, they form isocyanides or carbylamines, which have a foul odor. This reaction does not occur in secondary or tertiary amines.



31. The reagents that can be used to convert benzene diazonium chloride to benzene are

(A) $\text{SnCl}_2 / \text{HCl}$

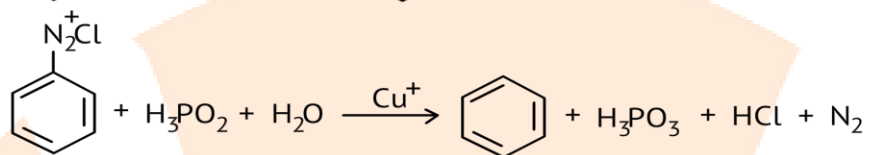
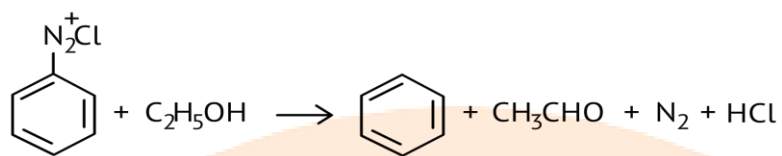
(B) $\text{CH}_3\text{CH}_2\text{OH}$

(C) H_3PO_2

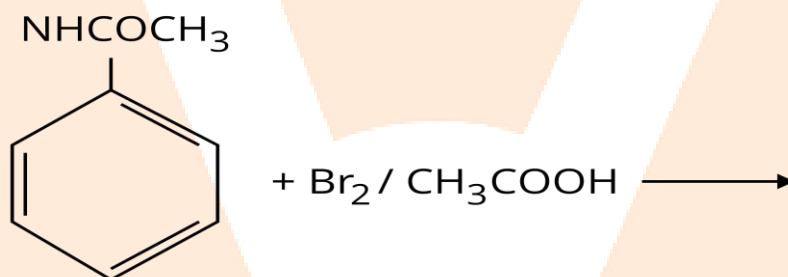
(D) LiAlH_4

Ans: Correct answer is (B and C)

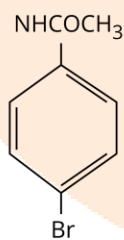
Certain mild reducing agents like hypophosphorous acid (phosphinic acid) or ethanol reduce diazonium salts to arenes and themselves get oxidised to phosphorous acid and ethanol, respectively.



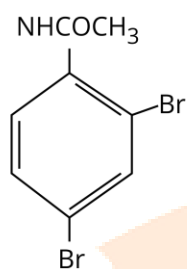
32. The Product of the following reaction is.



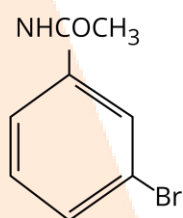
(A)



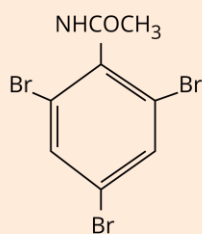
(B)



(C)

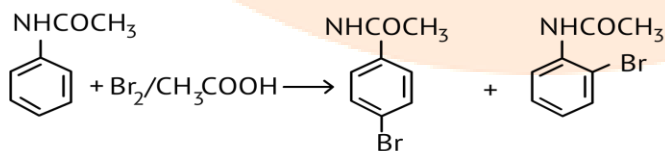


(D)



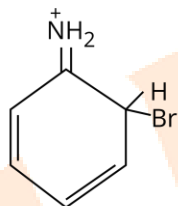
Ans: Correct answer is (A and B)

NHCOCH_3 is an ortho- para directing group so ortho-para products are formed.

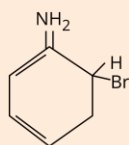


33. Arenium ion involved in the bromination of aniline is.

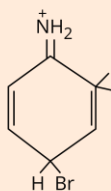
(A)



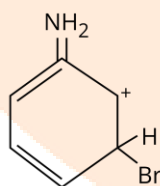
(B)



(C)

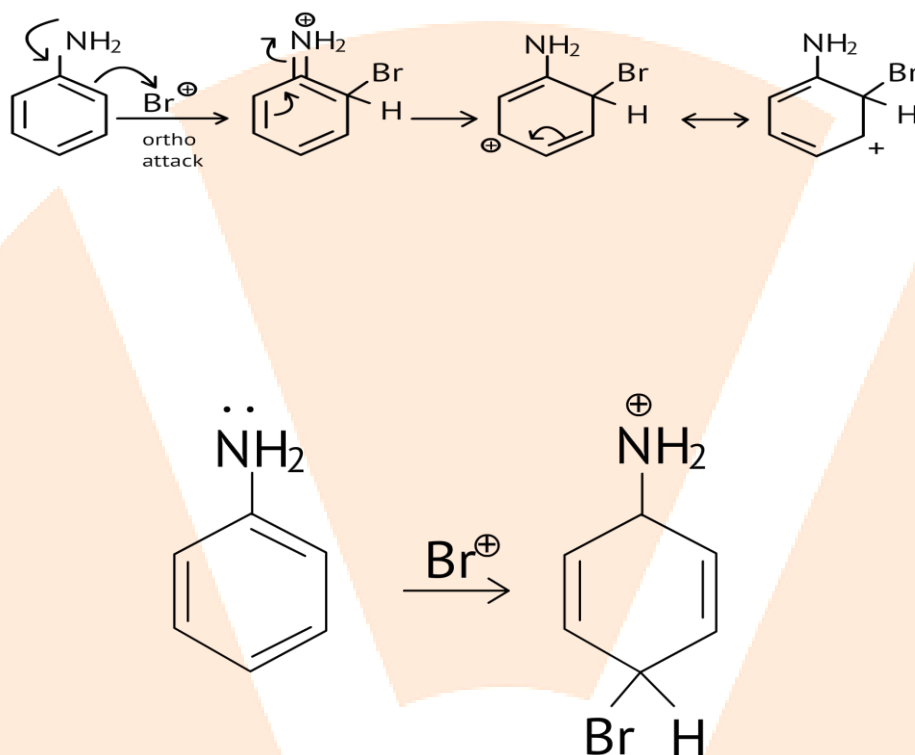


(D)



Ans: Correct answer is (A, B & C)

As the NH_2 group is Ortho para directing group, so Br^+ attacks only at the o and p positions thus the products formed corresponds to the option A, B and C.



34. Which of the following amines can be prepared by Gabriel synthesis?

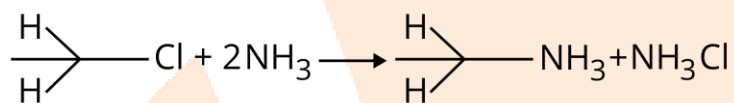
- (A) Isobutyl amine
- (B) 2-Phenylethylamine
- (C) N-Methylbenzylamine
- (D) Aniline

Ans: Correct answer is (A and B)

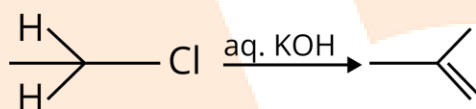
Gabriel synthesis is a method for producing primary amines. When phthalimide is treated with ethanolic potassium hydroxide, it forms a potassium salt of phthalimide, which when heated with an alkyl halide and then alkaline hydrolyzed yields the corresponding primary amine.

35. Which of the following reactions are correct?

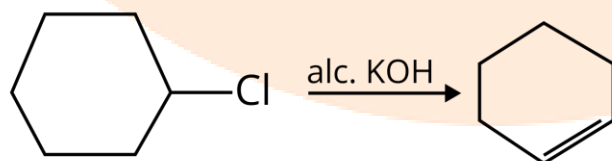
(A)



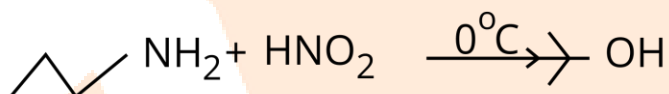
(B)



(C)



(D)



Ans: Correct answer is (A and C)

(a) Primary alkyl halides react with ammonia to give primary amines. Correct reaction.

(B) Elimination process doesn't occur with aq. KOH, hydrolysis process takes place. (B) is incorrect.

(c) Dehydrohalogenation i.e. elimination of HCl occurs which produces alkene as the product takes place with alc. KOH. (c) is correct.

(d) In this reaction, aliphatic primary amines, on treatment with nitrous acid, produce primary alcohol as a product.

Therefore (D) is incorrect.

36. Under which of the following reaction conditions, aniline gives p-nitro derivative as the major product?

(A) Acetyl chloride/pyridine followed by reaction with conc. H_2SO_4 + conc. HNO_3

(B) Acetic anhydride/pyridine followed by conc. H_2SO_4 + conc. HNO_3

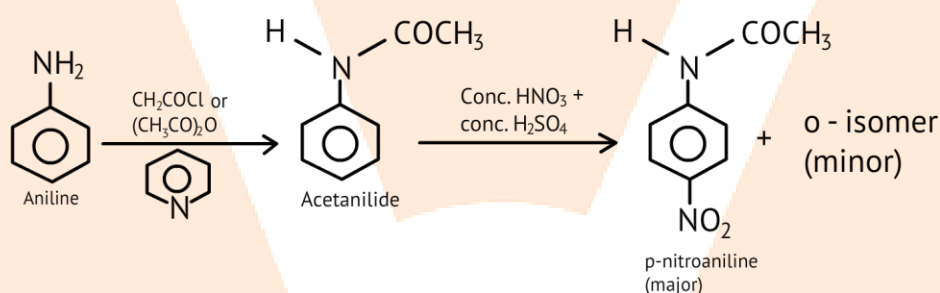
(C) Dil. HCl followed by reaction with conc. H_2SO_4 + conc. HNO_3

(D) Reaction with conc. H_2SO_4 + conc. HNO_3

Ans: Correct answer is (A and B)

In addition to the nitro derivatives, direct nitration of aniline produces tarry oxidation products. Furthermore, aniline is protonated in the strongly acidic medium to form the anilinium ion, which is meta directing. As a result, in addition to the ortho and para derivatives, a significant amount of meta derivative is formed.

However, by protecting the $-NH_2$ group with an acetylation reaction with acetic anhydride, the nitration reaction can be controlled and the p-nitro derivative obtained as the main product.

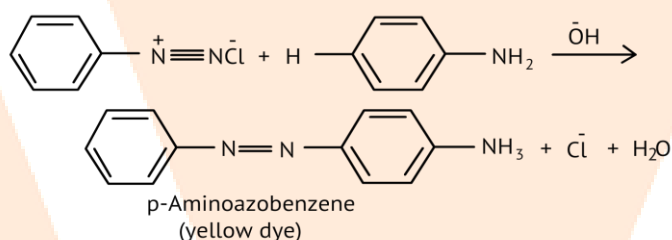


37. Which of the following reactions belong to electrophilic aromatic substitution?

- (A) Bromination of acetanilide
- (B) Coupling reaction of aryldiazonium salts
- (C) Diazotization of aniline
- (D) Acylation of aniline

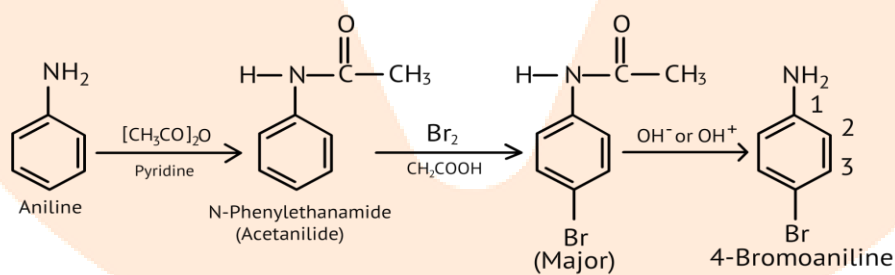
Ans: Correct answer is (A and B)

Benzene diazonium chloride reacts with phenol to form p-hydroxy azobenzene by coupling the phenol molecule at its para position with the diazonium salt. This is referred to as a coupling reaction. This is an illustration of an electrophilic substitution reaction.



At room temperature, aniline reacts with bromine water to form a white precipitate of 2,4,6-tribromoaniline.

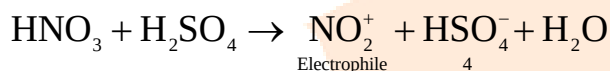
By acetylation with acetic anhydride to protect the $-\text{NH}_2$ group, then carrying out the desired substitution followed by hydrolysis of the substituted amide to the substituted amine.



Short Answer Type Question

38. Which is the role of HNO_3 in the nitrating mixture used for nitration of benzene?

Ans: In the nitrating mixture of conc. H_2SO_4 + conc. HNO_3 , HNO_3 act as base to generate the nitronium ion which is an initiation species for the nitration reaction.

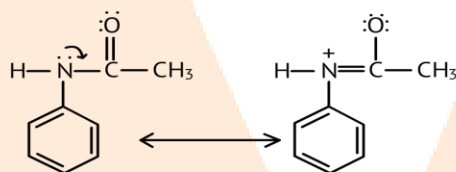


39. Why is $-\text{NH}_2$ group of aniline acetylated before carrying out nitration?

Ans: As $-\text{NH}_2$ is a strong activating group, the aniline will readily undergo electrophilic substitution reaction, and it is difficult to cease reaction at the mono substitution stage. So, direct acetylation of aniline is not possible as it undergoes oxidation.

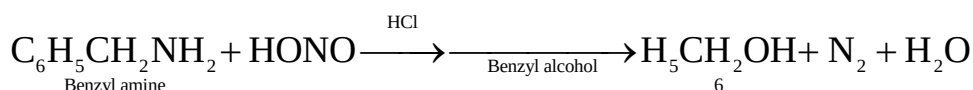
Therefore, the activating group $-\text{NH}_2$ is protected by an acetylation process and then followed by the nitration reaction to obtain a monosubstituted nitrated product.

The acetylated complex formed utilises the lone pair of nitrogen and is less available for donation, this helps to carry out the nitration reaction easily.



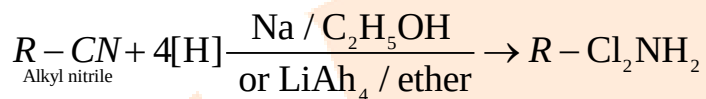
40. What is the product when $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ reacts with HNO_2 ?

Ans: When Benzyl amine is treated with nitrous acid, firstly BDC is formed which is unstable and decomposes to Benzyl alcohol.

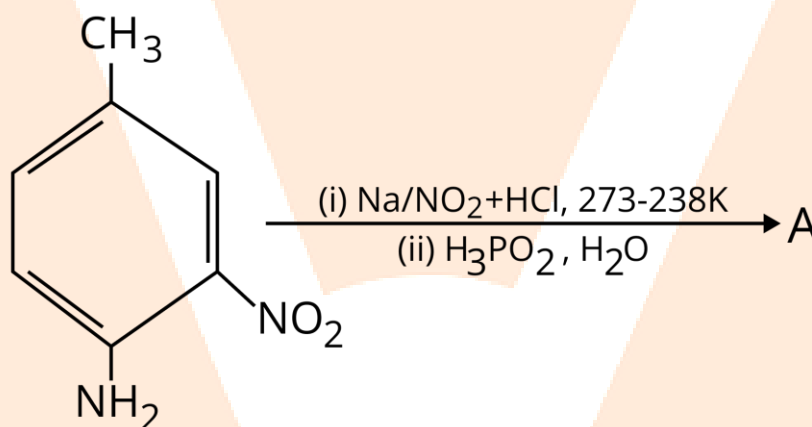


41. What is the best reagent to convert nitrile to primary amine?

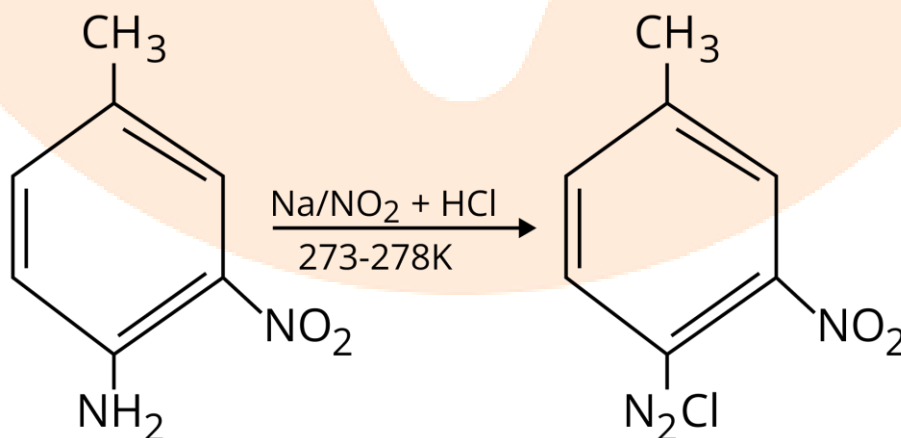
Ans: On treatment of nitriles with strong reducing agents such as lithium aluminium hydride in ether or sodium metal in alcohol, primary amines are obtained.

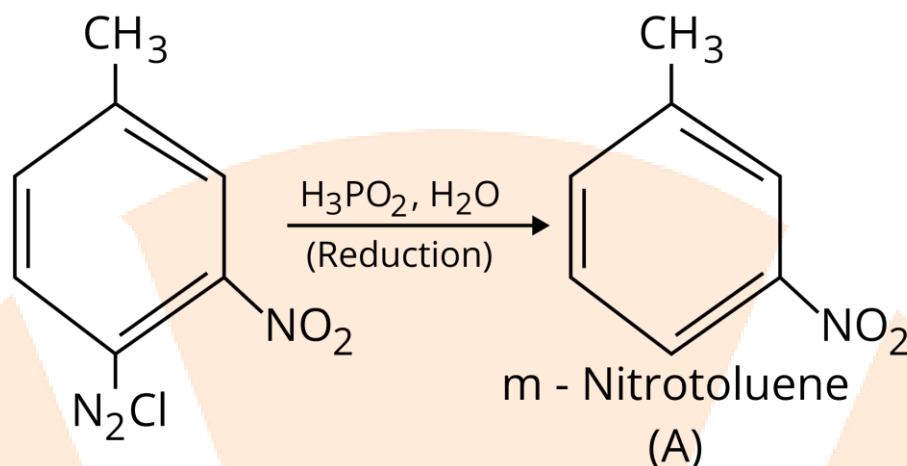


42. Give the structure of 'A' in the following reaction.



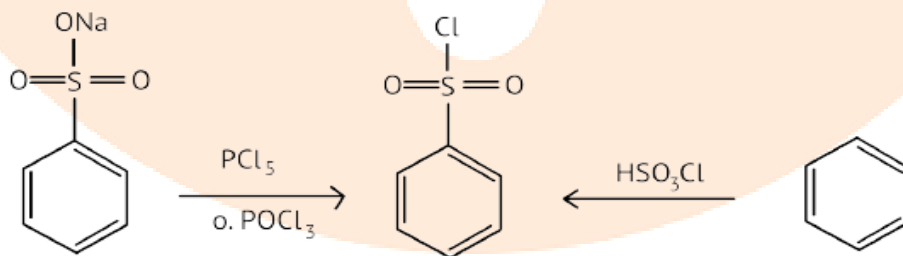
Ans:





43. What is Hinsberg reagent?

Ans: Hinsberg's reagent is also known as benzenesulphonyl chloride ($\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$). When it reacts with primary and secondary amines, it produces sulphonamides. Allowing secondary and tertiary amines to react with Hinsberg's reagent allows them to be distinguished (benzenesulphonyl chloride $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$). Secondary amines react with Hinsberg's reagent to form an alkali-insoluble product. N, N-diethylamine, for example, reacts with Hinsberg's reagent to form N,N-diethylbenzenesulphonamide, which is insoluble in alkalis. Tertiary amines, on the other hand, are unaffected by Hinsberg's reagent.



44. Why is benzene diazonium chloride not stored and is used immediately after its preparation?

Ans: The reaction of aniline with nitrous acid at 273-278 K produces benzene diazonium chloride. The reaction of sodium nitrite with hydrochloric acid produces nitrous acid in the reaction mixture. Diazotisation is the process of converting primary aromatic amines into diazonium salts. Because of its instability, the diazonium salt is generally not stored or used immediately after preparation.

The crystalline solid benzene diazonium chloride is colorless.

It is easily soluble in water and stable at room temperature, but it reacts with water when warmed.

In the dry state, it decomposes easily.

45. Why does acetylation of $-NH_2$ group of aniline reduce its activating effect?

Ans: As $-NH_2$ is a strong activating group, the aniline will readily undergo electrophilic substitution reaction, and it is difficult to cease reaction at the mono substitution stage.

Therefore, the activating group $-NH_2$ is protected by an acetylation process.

The acetylated complex formed utilizes the lone pair of nitrogen and are less available for donation, this helps to carry out the nitration reaction easily.

46. Explain why is $MeNH_2$ stronger base than $MeOH$?

Ans: As the electronegativity of oxygen is more than the electronegativity of a nitrogen atom, the $O-H$ bond is more polar than the $N-H$ bond, therefore $MeOH$ is stronger acid than $MeNH_2$ or $MeNH_2$ is stronger base than $MeOH$.

47. What is the role of pyridine in the acylation reaction of amines?

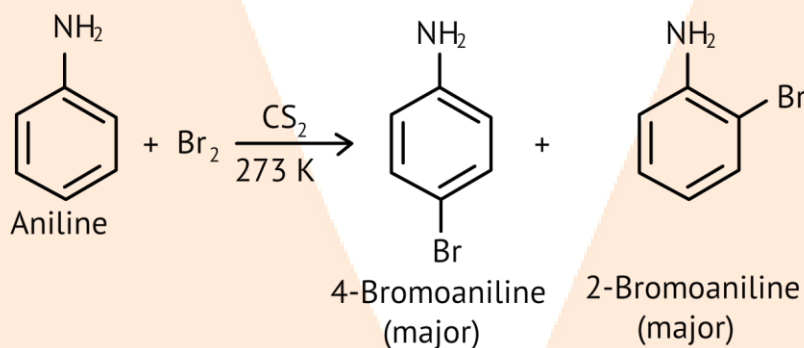
Ans: Amides are the byproducts of the acylation reaction. The reaction is carried out in the presence of a stronger base than the amine, such as pyridine, which removes the formed HCl and shifts the equilibrium to the right.

48. Under what reaction conditions (acidic/basic), the coupling reaction of aryl diazonium chloride with aniline is carried out?

Ans: The azo products have an extended conjugate system with both aromatic rings linked by the $-N=N-$ bond. These compounds are frequently colored and used as dyes. Benzene diazonium chloride reacts with phenol to form p-hydroxy azobenzene by coupling the phenol molecule in its para position with the diazonium salt. This is referred to as a coupling reaction.

49. Predict the product of reaction of aniline with bromine in non-polar solvent such as CS_2 .

Ans: CS_2 is a non-polar solvent which decreases the activating effect of $-NH_2$. As a result, mono substitution occurs only at o- and p- positions giving a mixture of 2-bromoaniline (minor) and 4-bromoaniline (major) as products.



50. Arrange the following compounds in increasing order of dipole moment:

$CH_3CH_2CH_3, CH_3CH_2NH_2, CH_3CH_2OH$.

Ans: $CH_3CH_2CH_3 < CH_3CH_2NH_2 < CH_3CH_2OH$

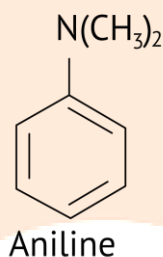
As oxygen is more electronegative than nitrogen therefore, the O-H bond is more polar than the N-H bond. So ethanol has more dipole moments than ethylamine. Propane is non- polar in nature hence, had the least among all.

51. What is the structure and IUPAC name of the compound, allyl amine?

Ans: Prop-2-en-1-amine



52. Write down the IUPAC name of



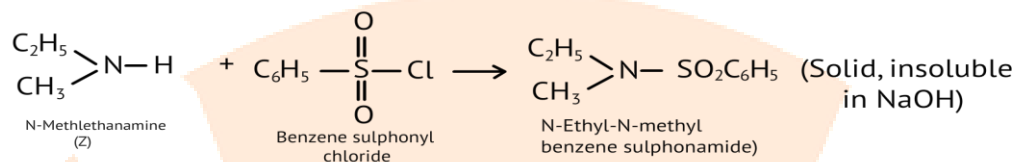
Ans: N, N-dimethylbenzenamine

53. A compound Z with molecular formula $\text{C}_3\text{H}_9\text{N}$ reacts with $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ to give a solid, insoluble in alkali, identify Z.

Ans: Z is an aliphatic amine that produces a solid that is insoluble in base. This means that the reaction with $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ must produce a product with no replaceable hydrogen attached to nitrogen.

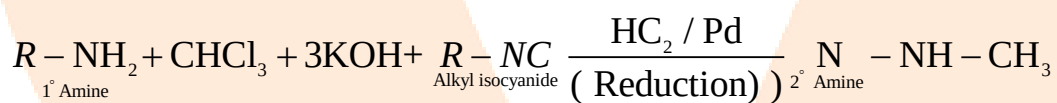
To put it another way, the amine must be a secondary amine.

Z stands for ethylmethanamine.

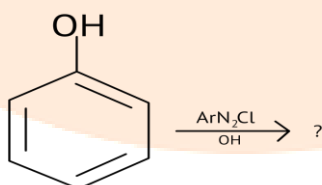


54. A primary amine, RNH_2 can be reacted with $\text{CH}_3 - \text{X}$ to get secondary amine, $\text{R} - \text{NHCH}_3$ but the only disadvantage is that 3° amine and quaternary ammonium salts are also obtained as side products. Can you suggest a method where RNH_2 forms only 2° amine?

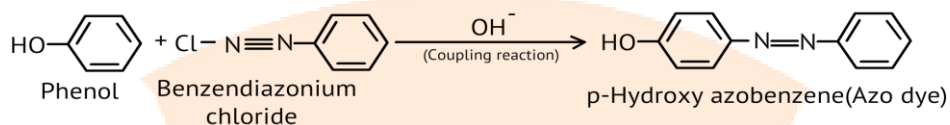
Ans: Carbylamine reaction is shown by primary amines only, primary amines react with chloroform in the presence of alcoholic KOH to form isocyanides which upon catalytic reduction gives secondary amines as the final product.



55. Complete the following reaction.

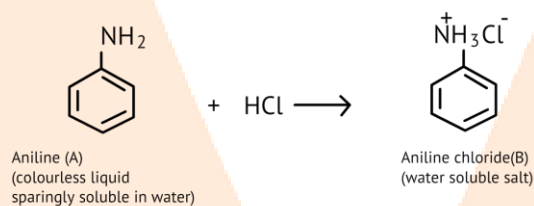


Ans:

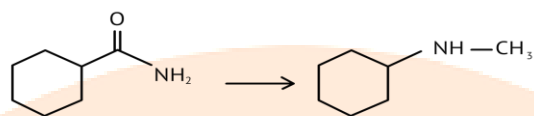


56. Why is aniline soluble in aqueous HCl?

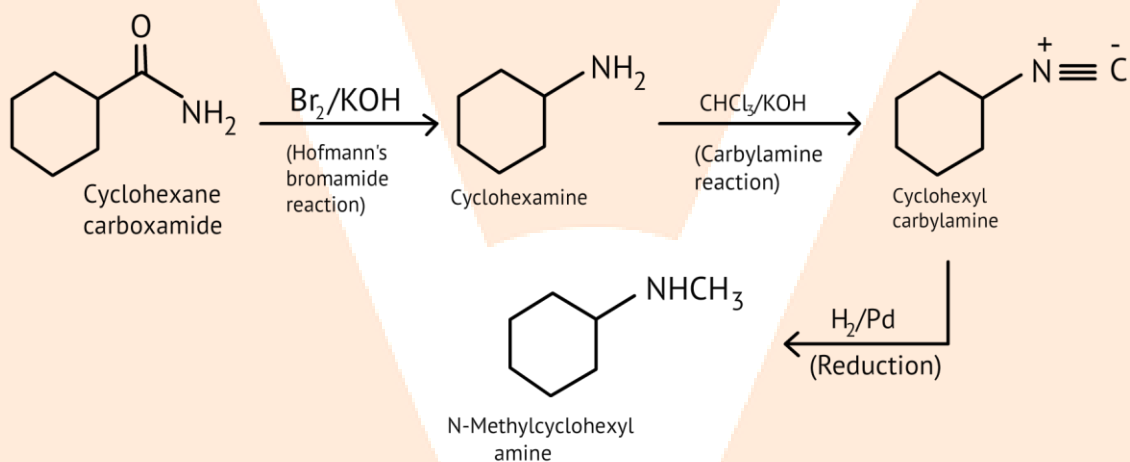
Ans: On reaction with hydrochloric acid, aniline forms anilinium chloride ion which is water soluble. Therefore, the aniline solubility in aqueous HCl solution.



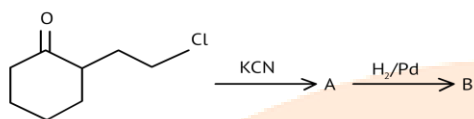
57. Suggest a route by which the following conversion can be accomplished.



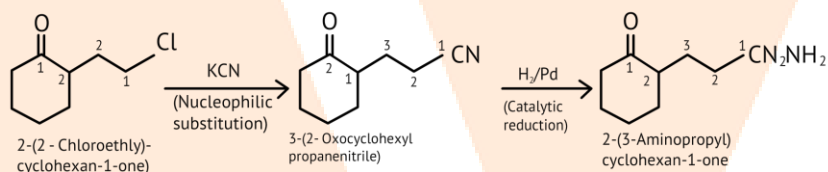
Ans:



58. Identify A and B in the following reaction.



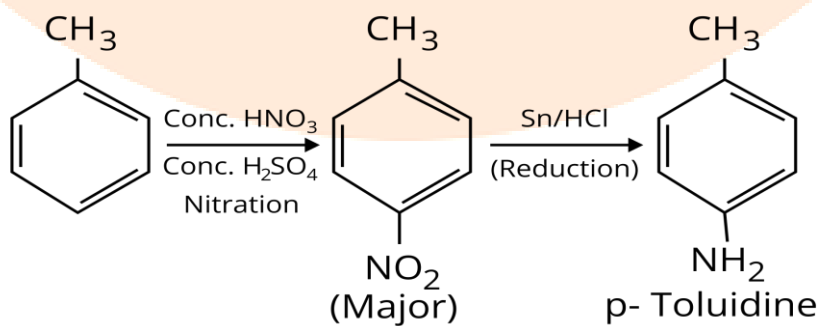
Ans:



59. How will you carry out the following conversions?

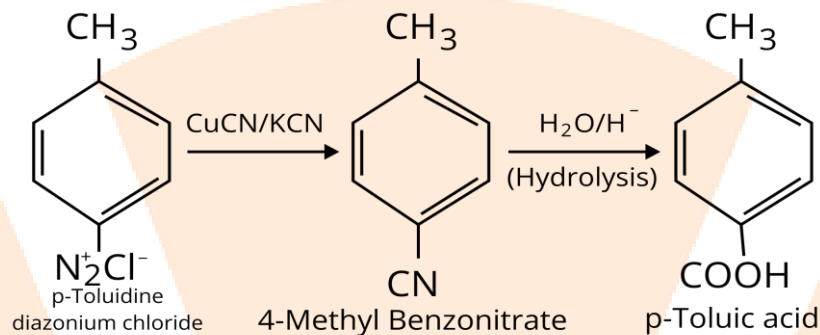
(i) toluene $\rightarrow$ p-toluidine

Ans:



(ii) p-toluidine diazonium chloride $\rightarrow$ p-toluic acid

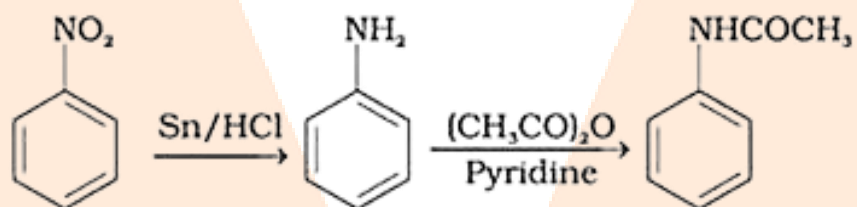
Ans:



60. Write following conversions:

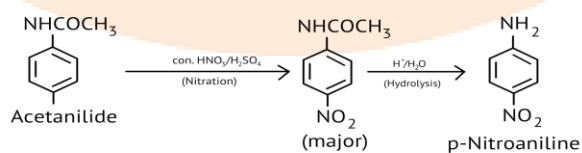
(i) nitrobenzene $\rightarrow$ acetanilide

Ans:



(ii) acetanilide $\rightarrow$ p-nitroaniline

Ans:



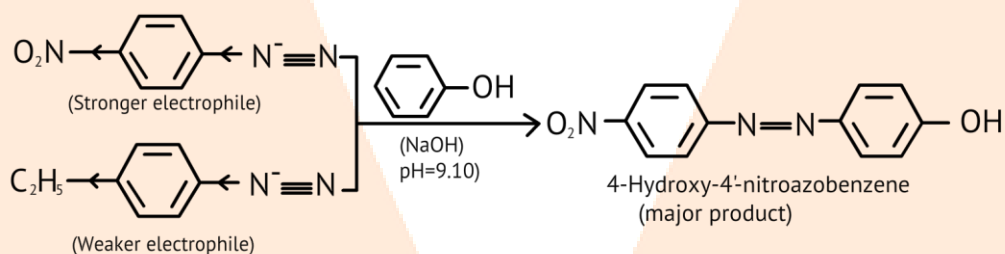
61. A solution contains 1 g mol each of p-toluene diazonium chloride and p-nitrophenyl diazonium chloride. To this 1 g mol of alkaline solution of phenol is added. Predict the major product. Explain your answer.

Ans: This is an example of an electrophilic aromatic substitution reaction. phenol generates phenoxide ion in alkaline medium, which is more electron-rich than phenol and thus more reactive to electrophilic attack. In this reaction, the electrophile is an aryldiazonium cation.

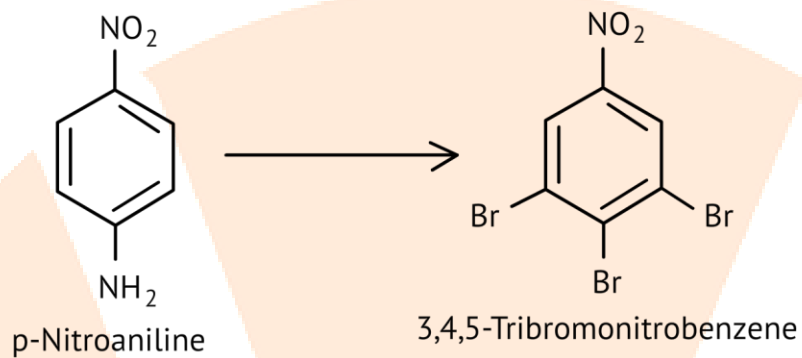
The faster the reaction, the stronger the electrophile.

The cation p-nitrophenyldiazonium is a stronger electrophile than the cation p-toluene diazonium.

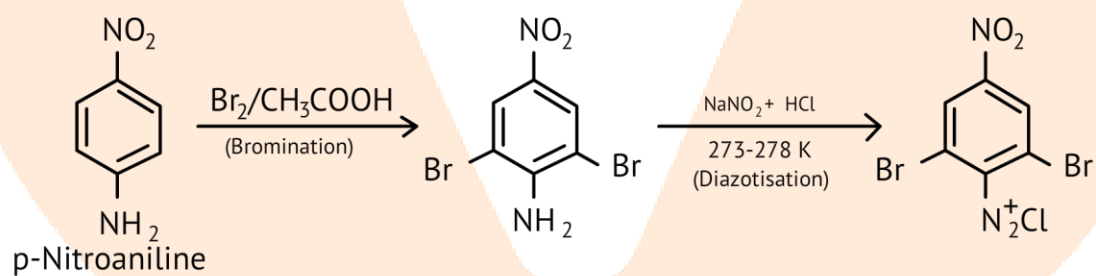
As a result, it preferentially couples with phenol.

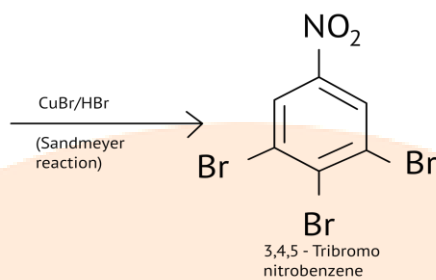


62. How will you bring out the following conversion?

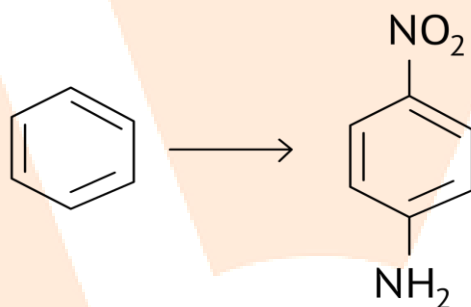


Ans:

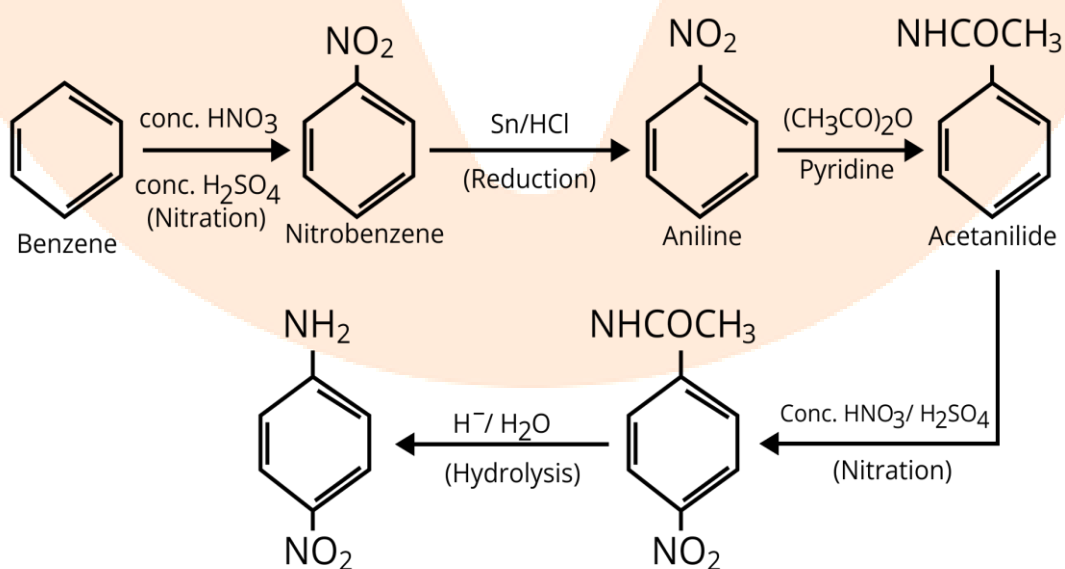




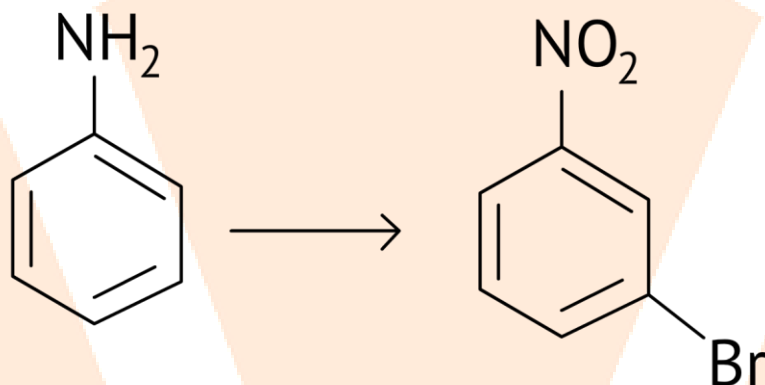
63. How will you carry out the following conversion?



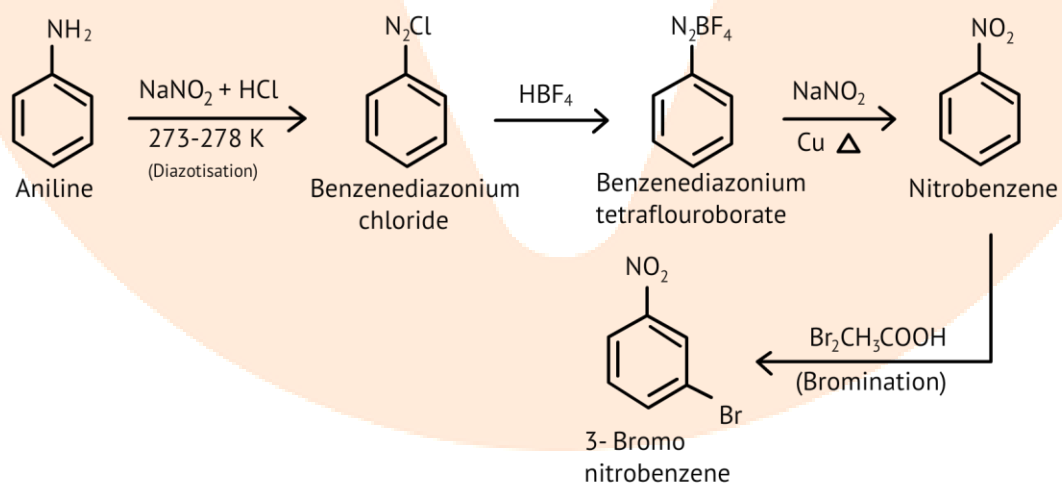
Ans:



64. How will you carry out the following conversion?

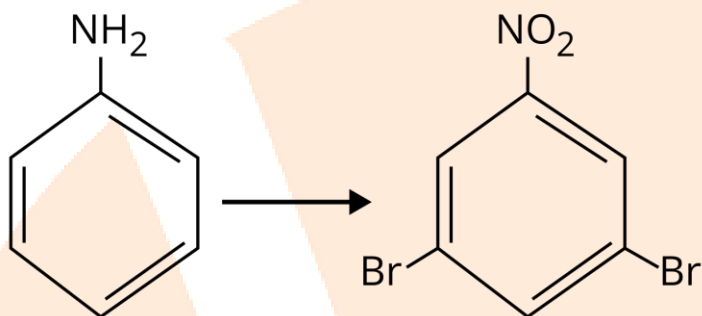


Ans:

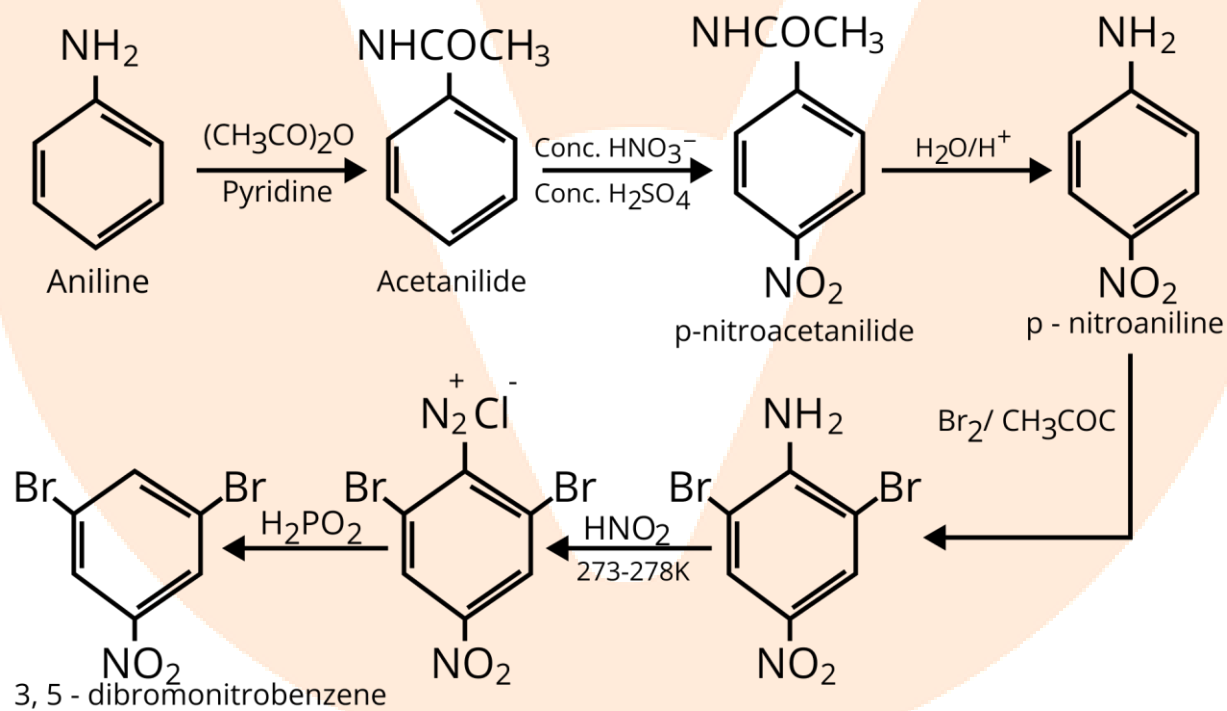


65. How will you carry out the following conversion?

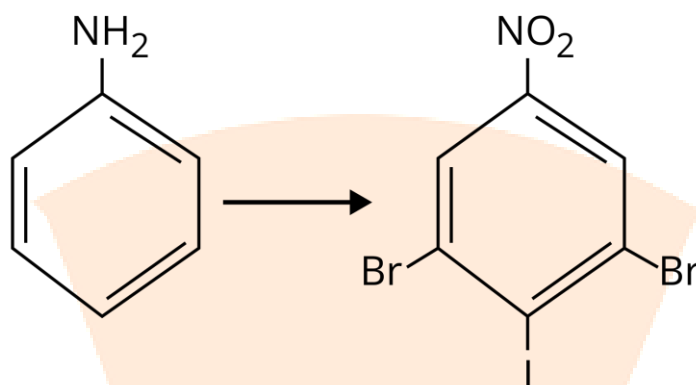
(i)



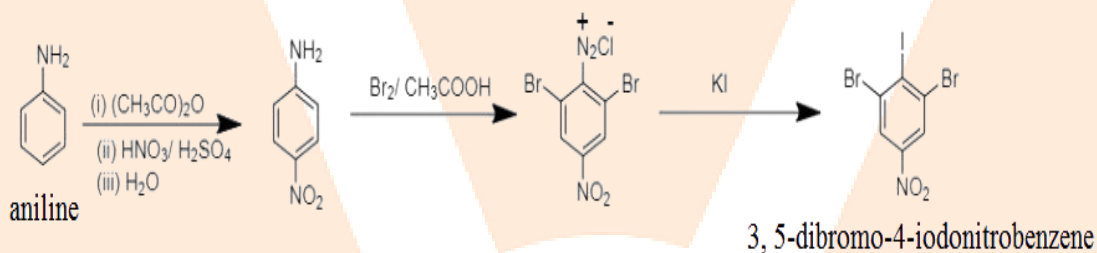
Ans:



(ii)



Ans:



Matching Type Question

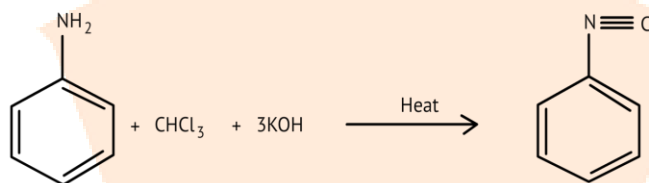
66. Match the reactions given in Column I with the statements given in Column II.

Column I	Column II
(i) Ammonolysis	(a) Amine with lesser number of carbon atoms
(ii) Gabriel phthalimide synthesis	(B) Detection test for primary amines.
(iii) Hoffmann Bromide reaction	(c) Reaction of phthalimide with KOH and R-X
(iv) Carbylamine reaction	(d) Reaction of alkylhalides with NH_3

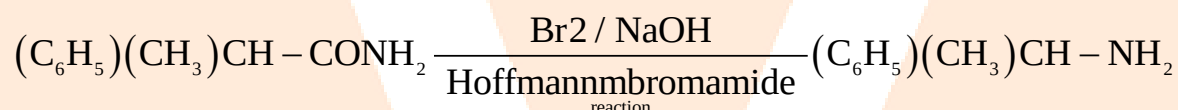
Ans: (i)-(d), (ii)-(c), (iii)-(a), (iv)-(B)

Ammonolysis: $\text{R} - \text{X} + \text{NH}_3 \rightarrow \text{RNH}_2 + \text{HCl}$

Carbylamine reaction:



Hoffmann bromamide reaction:



67. Match the compounds given in Column I with the items given in Column II.

Column I	Column II
(i) Benzene sulphonyl chloride	(a) Zwitter ion
(ii) Sulphanilic acid	(B) Hinsberg reagent
(iii) Alkyl diazonium salts	(c) Dyes
(iv) Aryl diazonium salts	(d) Conversion to alcohols

Ans: Correct answer is

Column I	Column II
(i) Benzene sulphonyl chloride	(B) Hinsberg reagent
(ii) Sulphanilic acid	(a) Zwitter ion
(iii) Alkyl diazonium salts	(d) Conversion to alcohols
(iv) Aryl diazonium salts	(c) Dyes

Assertion Reason Type

In the following questions, a statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices:

- (a) Both Assertion and Reason are wrong.**
- (B) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.**
- (c) Assertion is correct but Reason is wrong.**
- (d) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.**
- (e) Assertion is wrong but Reason is correct.**

68. Assertion (A): Acylation of amines gives a monosubstituted product, whereas alkylation of amines gives polysubstituted product.

Reason (R): Acyl group sterically hinders the approach of further acyl groups.

Ans: Correct answer is (c)

In the acyl derivative, the delocalisation of electrons of the nitrogen atom occur over the carbonyl group, this decreases the electron density on the the nitrogen atom that it no more perform as nucleophile and don't react with next acylating alkaline molecule.

Therefore, Assertion statement is correct but reason is incorrect.

69. Assertion (A): Hofmann's bromamide reaction is given by primary amines.

Reason (R): Primary amines are more basic than secondary amines.

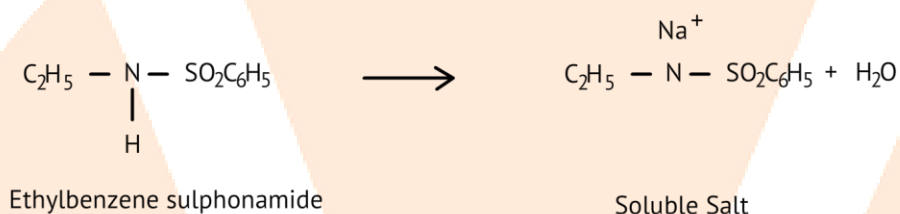
Ans: Correct answer is (a)

In Hoffmann bromamide degradation reaction as in this reaction primary amide group is treated with halogen first bromine then the halogen substituted amide product is coverts to primary amine with the release of carbon dioxide gas. Both the statements assertion and reason are incorrect therefore the correct option is a.

70. Assertion (A): N-Ethylbenzene sulphonamide is soluble in alkali.

Reason (R): Hydrogen attached to nitrogen in sulphonamide is strongly acidic.

Ans: Correct answer is (d)



Ethylbenzene sulphonamide is soluble in alkali because it has acidic hydrogen.

71. Assertion (A): N, N-diethylbenzene sulphonamide is insoluble in alkali.

Reason (R): Sulphonyl group attached to nitrogen atom is strong electron withdrawing group.

Ans: Correct answer is (b)

N, N-diethylbenzene sulphonamide, there is no acidic hydrogen present on the N-atom which can make it soluble. Therefore, it is insoluble in alkali.

72. Assertion (A): Only a small amount of HCl is required in the reduction of nitro compounds with iron scrap and HCl in the presence of steam.

Reason (R): FeCl₂ formed gets hydrolysed to release HCl during the reaction.

Ans: Correct answer is (d)



The nascent hydrogen formed act as reducing agent for the reduction of nitro compounds.

73. Assertion (A): Aromatic 1° amines can be prepared by Gabriel phthalimide synthesis.

Reason (R): Aryl halides undergo nucleophilic substitution with anion formed by phthalimide.

Ans: Correct answer is (a)

Aromatic primary amines cannot be prepared using Gabriel phthalimide synthesis as aryl halides do not undergo nucleophilic substitution with anion formed by phthalimide. Both the statement's assertion and reason are incorrect, hence the Ans: Correct answer is a.

74. Assertion (A): Acetanilide is less basic than aniline.

Reason (R): Acetylation of aniline results in decrease of electron density on nitrogen.

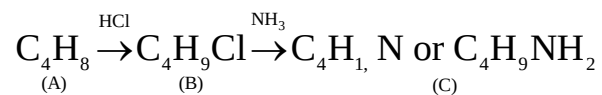
Ans: Correct answer is (d)

Acetanilide is less basic than aniline because electron density of nitrogen is lowered by acetyl group.

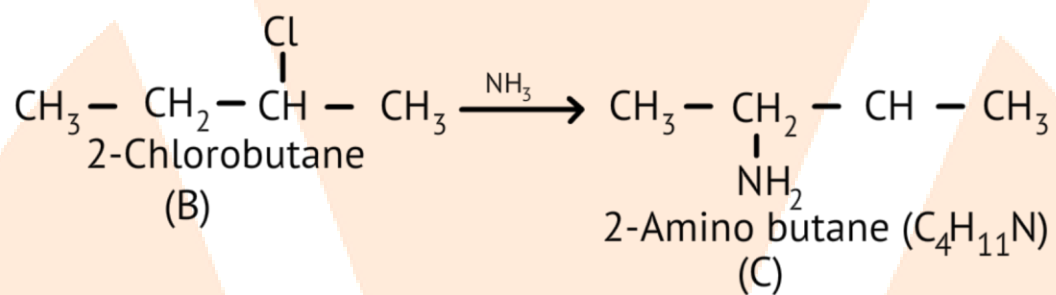
Long Answer Type Question

75. A hydrocarbon ' A ', (C_4H_8) on reaction with HCl gives a compound ' B ' ($\text{C}_4\text{H}_9\text{Cl}$), which on reaction with 1mol of NH_3 gives compound ' C ', ($\text{C}_4\text{H}_{11}\text{N}$). On reacting with NaNO_2 and HCl followed by treatment with water, compound ' C ' yields an optically active alcohol, ' D '. Ozonolysis of ' A ' gives 2 moles of acetaldehyde. Identify compounds 'A' to 'D'. Explain the reactions involved.

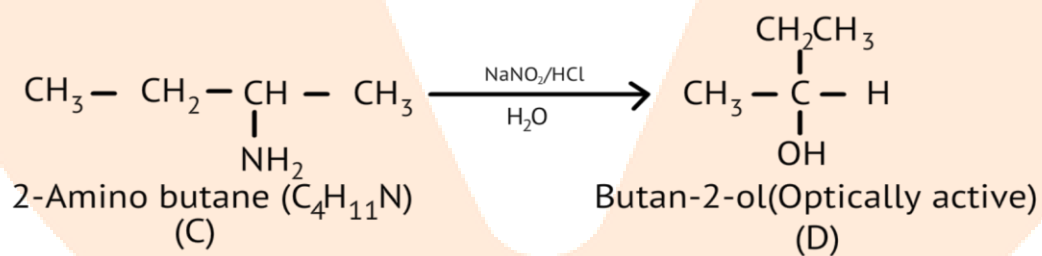
Ans:



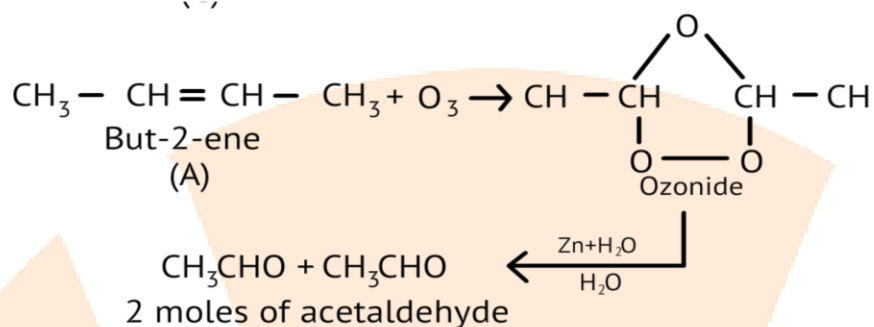
(ii)



(iii)



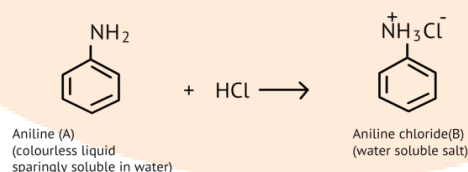
(iv)



76. A colourless substance 'A' ($\text{C}_6\text{H}_7\text{N}$) is sparingly soluble in water and gives a water soluble compound 'B' on treatment with mineral acid. On reacting with CHCl_3 and alcoholic potash 'A' produces an obnoxious smell due to the formation of compound 'C'. Reaction of 'A' with benzenesulphonyl chloride gives compound 'D' which is soluble in alkali. With NaNO_2 and, 'A' forms compound 'E' which reacts with phenol in alkaline medium to give an orange dye 'F'. Identify compounds 'A' to 'F'.

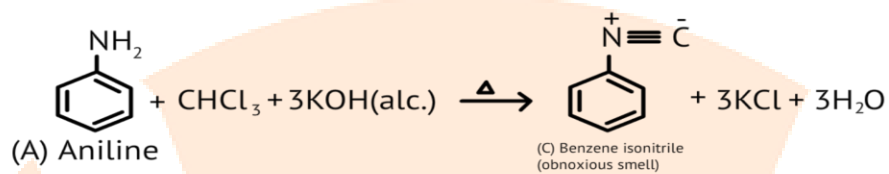
Ans:

(i)

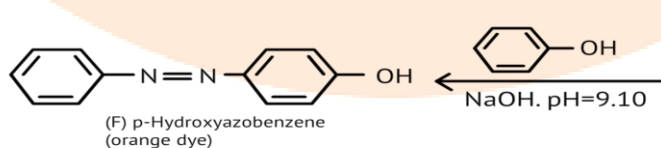
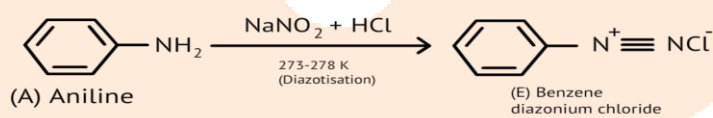
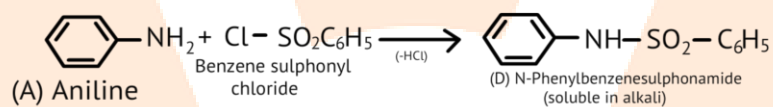


(ii)

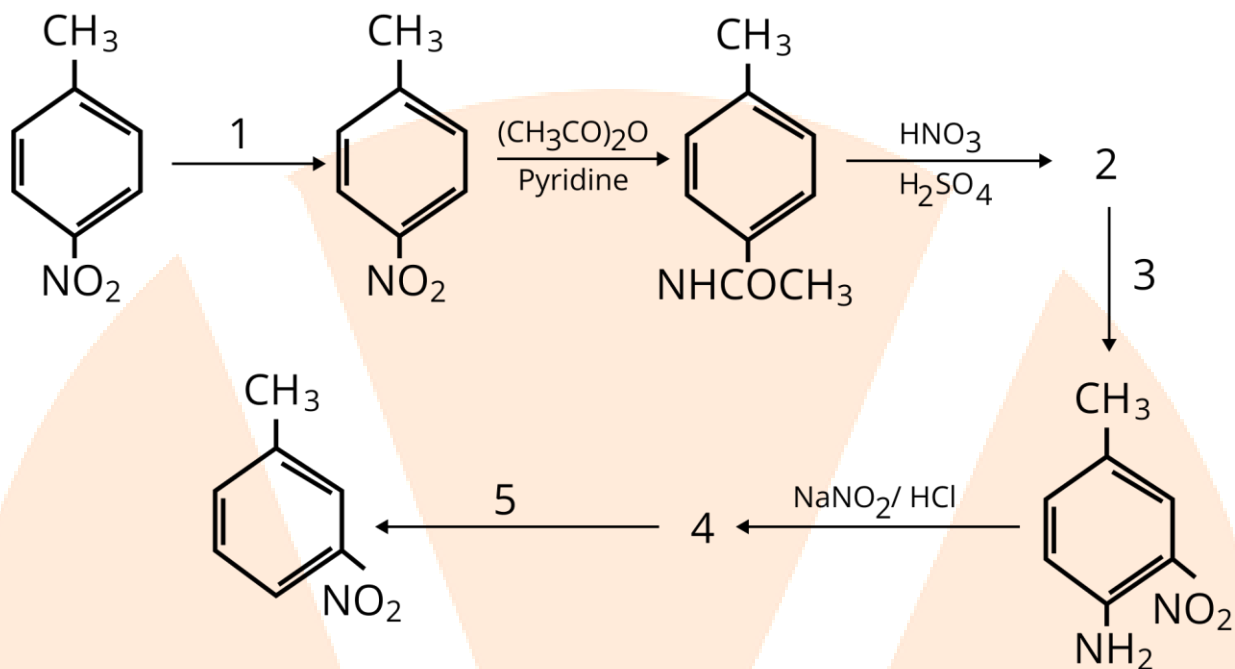
(iii)



(iv)



77. Predict the reagent or the product in the following reaction sequence.



Ans:

